

# **Air Quality Classification Report**

## 1. Introduction

Air pollution is one of the major environmental challenges affecting public health, especially in large urban areas such as Delhi NCR.

The objective of this project is to build a machine learning model to classify Air Quality Index (AQI) levels based on environmental and pollutant data..

## 2. Dataset Description

- The dataset used in this project is the **Delhi Pollution AQI Dataset** obtained from Kaggle.
- It contains air quality and environmental measurements collected from multiple monitoring stations across the Delhi NCR region during the period (2020–2025).

### Dataset Features:

- **Pollutants:** PM2.5, PM10, CO, NO2, SO2, O3
- **Weather Features:** Temperature, Wind Speed, Visibility, day\_of\_week

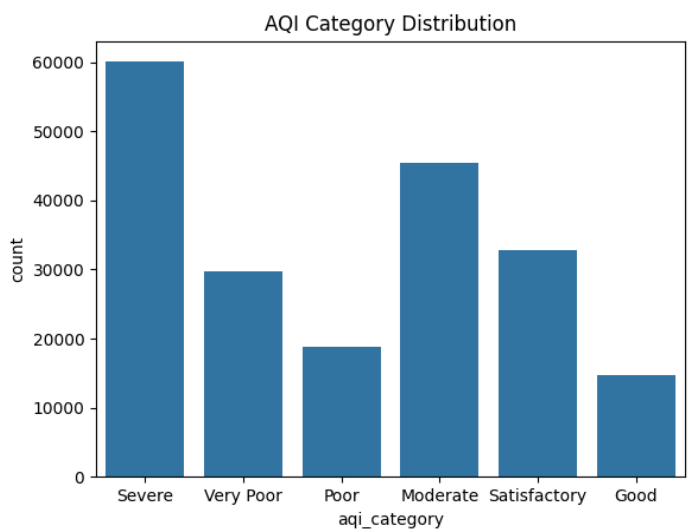
### Target Variable:

- AQI Category (Good, Moderate, Poor, Severe)

## 3. Exploratory Data Analysis (EDA)

- **Missing Values Check** The dataset was checked for missing values to ensure data quality
- **Class Distribution**

Figure 1: AQI Category Distribution



**Description:**

This figure shows the distribution of AQI categories in the dataset. It helps identify whether the dataset is balanced or imbalanced across different air quality levels

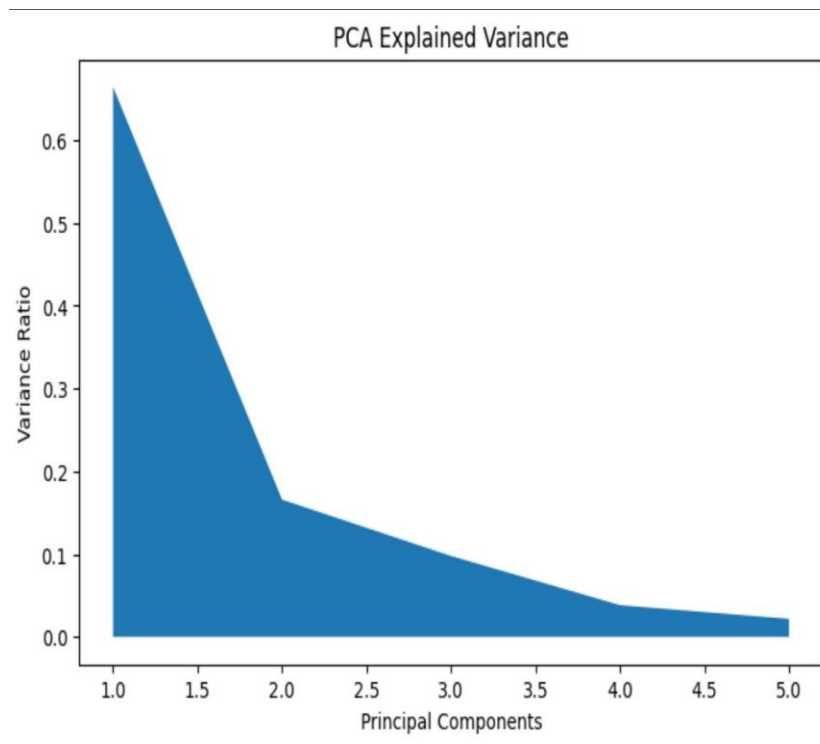
#### 4. Data Preprocessing

The dataset was preprocessed through several steps to ensure its suitability for machine learning models:

- Removal of irrelevant columns such as **datetime**, **date**, and **aqi**
- Encoding categorical variables using **Label Encoding**
- Normalization of numerical features using **StandardScaler**
- Splitting the dataset into **training** and **testing** sets (80% training, 20% testing) using stratified sampling

#### 5. Dimensionality Reduction

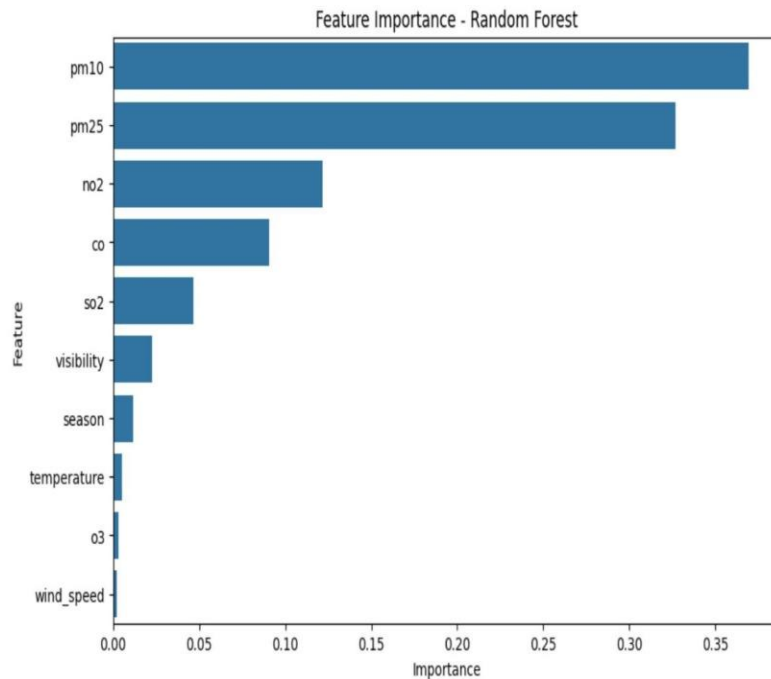
Figure 2: PCA Explained Variance

**Description:**

This figure shows how much variance is explained by each principal component. It helps in reducing dimensionality while preserving important information

## 6.Feature Selection

Figure 3: Feature Importance (Random Forest)



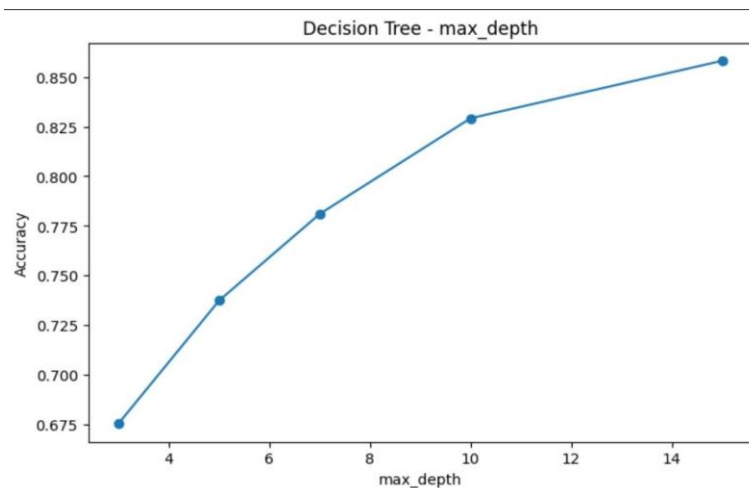
### Description:

This figure shows the importance of each feature in predicting AQI. Pollutants like PM2.5 and PM10 have the highest impact

## 7. Machine Learning Models

### 7.1 Decision Tree

Figure 4: Decision Tree – Max Depth vs Accuracy

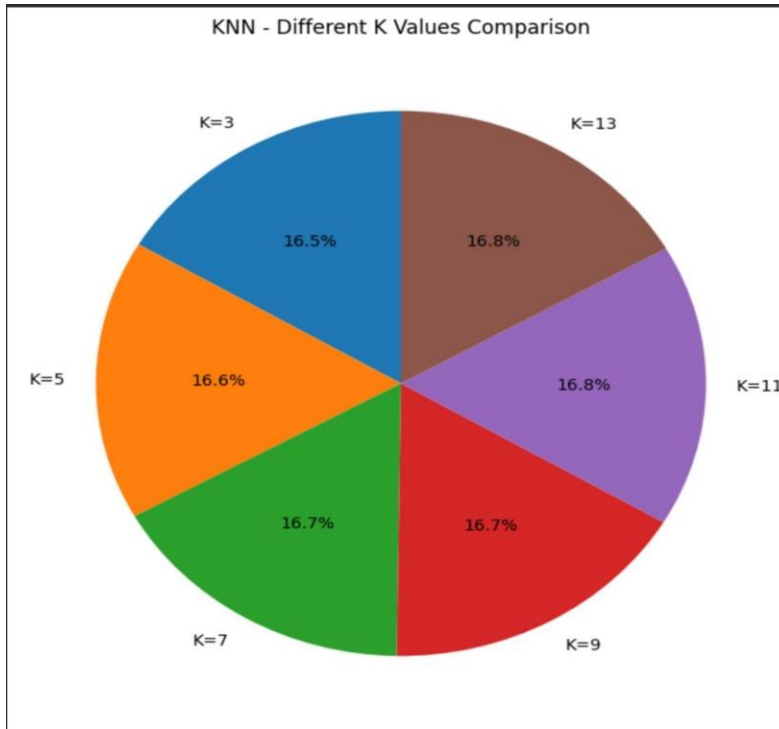


**Description:**

Shows how changing tree depth affects model performance and helps control overfitting

**7.2 K-Nearest Neighbors (KNN)**

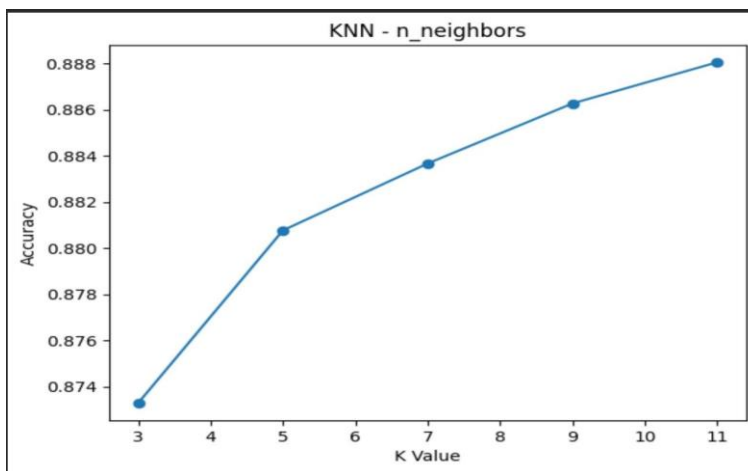
Figure 5: **KNN – Different K Values Comparison**



**Description:**

Compares performance across different K values.

Figure 6: **KNN – Optimal Number of Neighbors (K)**

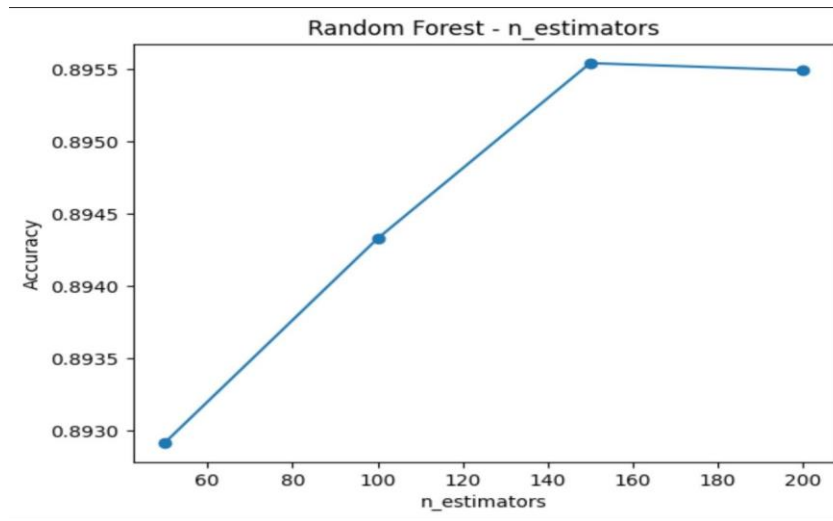


**Description:**

Shows relationship between K value and model accuracy

### 7.3 Random Forest

Figure 7: Random Forest – Number of Estimators vs Accuracy

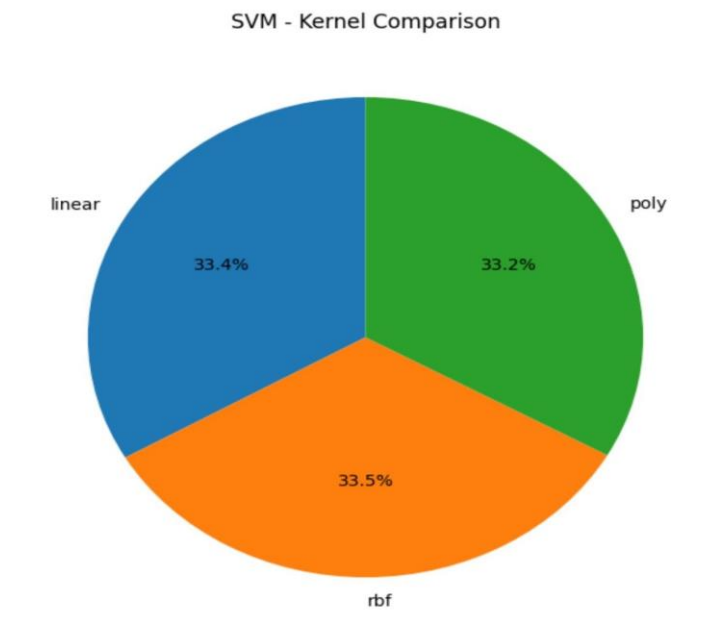


#### Description:

Shows how increasing number of trees improves model performance until it stabilizes

### 7.4 Support Vector Machine (SVM)

Figure 8: SVM Kernel Comparison



#### Description:

This figure illustrates the comparison between different SVM kernels: Linear, RBF, and Polynomial.

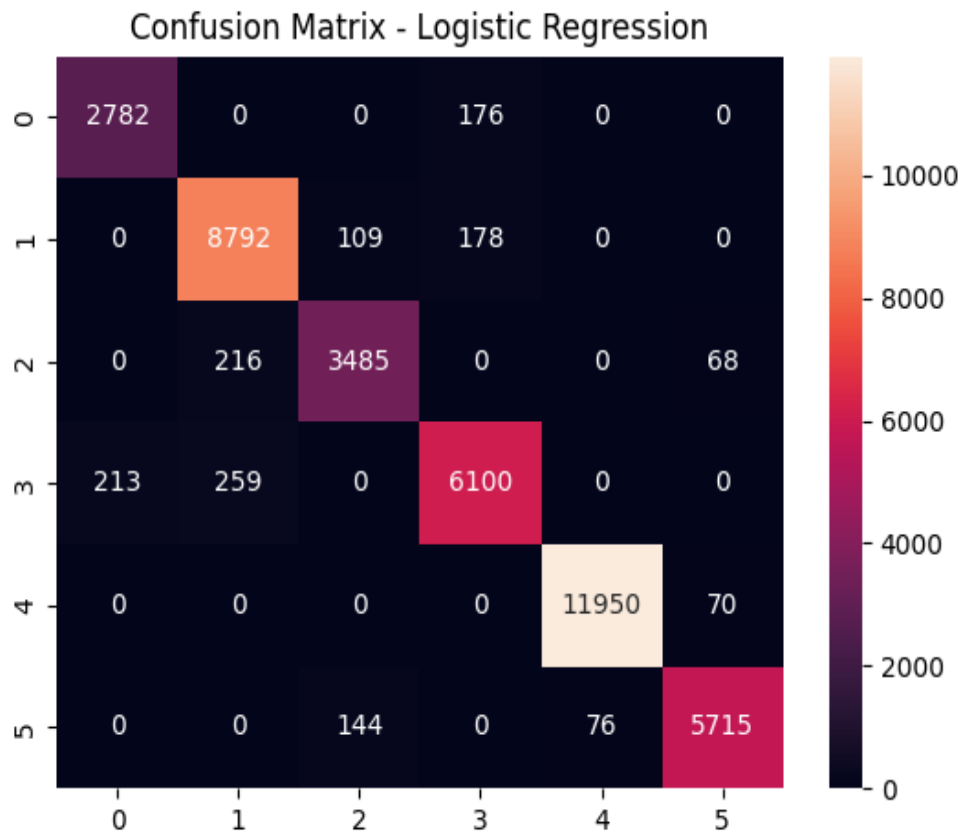
## 8. Evaluation Metrics

The models were evaluated using the following metrics:

- Accuracy Score
- Classification Report (Precision, Recall, F1-score)
- Confusion Matrix

### Confusion Matrices

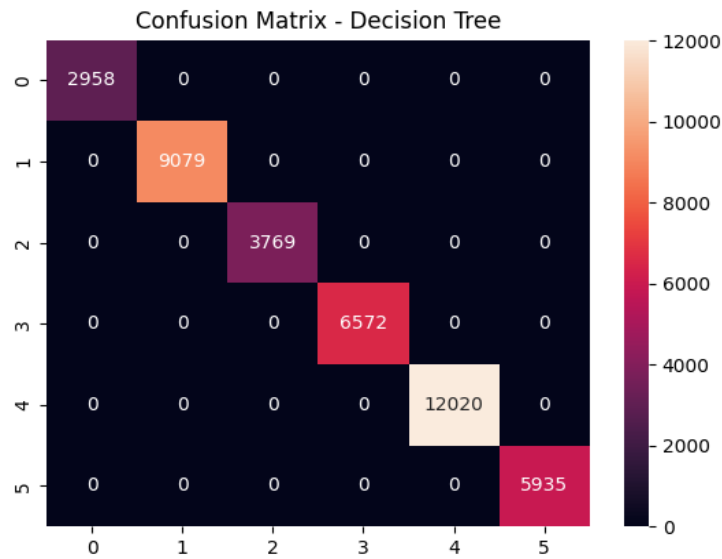
Figure 9: **Confusion Matrix – Logistic Regression**



#### Description:

This confusion matrix shows the performance of the Logistic Regression model by comparing predicted vs actual AQI categories

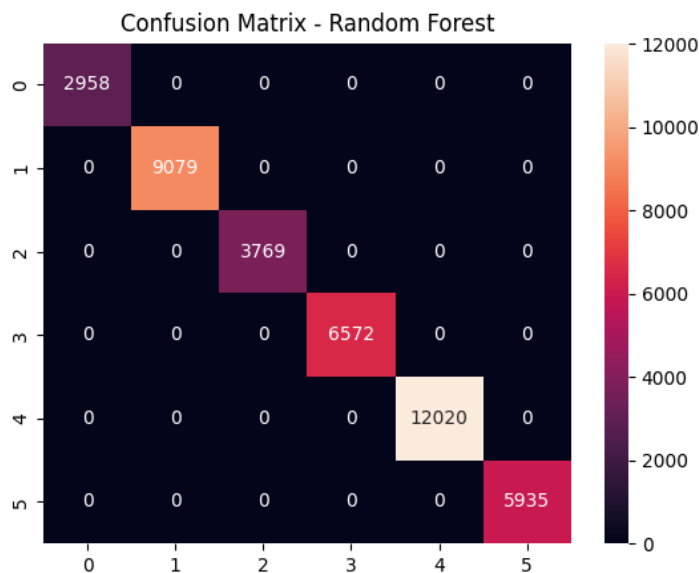
Figure 10: **Confusion Matrix -Decision Tree**



**Description:**

This confusion matrix illustrates the classification results of the Decision Tree model and highlights correctly and incorrectly classified samples.

Figure 11: **Confusion Matrix – Random Forest**



**Description:**

This confusion matrix shows the performance of the Random Forest model, which achieves better classification accuracy compared to other models.



## 9. Model Comparison

Figure 12: **Model Accuracy Comparison**



### Description:

This figure compares the accuracy of all models

### Results:

- Logistic Regression → baseline performance
- Decision Tree → improved after tuning
- Random Forest → best performance

## 10. Results and Discussion

The experimental results showed that:

- Random Forest achieved the best performance
  - Decision Tree showed moderate improvement
  - Logistic Regression provided baseline model.
- 
- ▶ Feature selection contributed to improving model efficiency and reducing noise.
  - ▶ A confusion matrix visualization was used to analyze classification errors and model performance.
  - ▶ Model comparison was performed using a bar chart to compare accuracy across models.

## 11. Conclusion

This project successfully implemented a machine learning pipeline for air quality classification.

### Key Findings:

- Feature selection improved model performance
  - Ensemble methods outperformed single models
  - Random Forest was the most effective model
- This demonstrates the effectiveness of machine learning in environmental monitoring and prediction tasks.

This is the project on Google Colab:

<https://colab.research.google.com/drive/1SH--1s6SK8-dD1MNRYNsmQxfJmcfHvow>